Library Management System (LiberSys)

System Architecture Specification & Technical Documentation — Category: Enterprise Full Stack Information Systems

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Repository: https://github.com/rakshapatil526/Library-Management-System

Status: Verified Repository

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1. Executive Summary & Problem Statement

LiberSys is a full-stack academic library information system built with React, Django REST Framework, and PostgreSQL. It streamlines the lifecycle of bibliographic assets and student borrowing operations, eliminating physical ledger errors, inventory discrepancies, and overdue tracking overhead.

The system enforces strict book availability tracking, student borrowing limits, and automated overdue penalty calculations.

2. Database Design & Relational Tables

Table	Primary Key	Schema Specifications
books	book_id (AutoField)	title (varchar 255), author, ISBN (unique), category, quantity, available_quantity
students	student_id (CharField)	name, email (unique), phone, department, enrollment_date
borrow_records	record_id (AutoField)	student_id (FK: students), book_id (FK: books), issue_date, due_date, return_date, status

3. Business Rules & Transaction Workflow

```
[Borrow Request] ■■> [Check Student Active Records (< Limit)]
                              ■
                              ▼ (Eligible)
                     [Check Book available_quantity > 0]
                              ■
                              ▼ (In Stock)
                     [Atomic DB Transaction:
                        1. Create borrow_record (status='ISSUED', due_date = today + 14 days)
                        2. Decrement available_quantity by 1]
                              ■
                              ▼
                     [Return Request:
                        1. Set return_date = today
                        2. Calculate Overdue Fine if return_date > due_date
                        3. Increment available_quantity by 1
                        4. Set status='RETURNED']
```

4. Technology Stack & Verification Status

- **Frontend:** React 18, Vite, Tailwind CSS, Lucide React Icons.
- **Backend:** Python 3.13, Django 5.2, Django REST Framework, django-cors-headers.
- **Database:** PostgreSQL with automatic local SQLite development fallback.
- **Status:** Fully verified from repository source code and database migration schemas.